

# OpsBrain AI

Unified Asset & Operations Brain for Zero-Harm Plants

**ET AI Hackathon 2026 | Problem Statement #8**

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Submission Version: 1.2 (Final Freeze)

## 2. Problem Statement Alignment

OpsBrain AI is engineered specifically for the ET AI Hackathon 2026 Problem Statement #8: "AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain."

### The Problem of Fragmented Silos:

In heavy industrial plants, crucial asset information is scattered across disconnected, incompatible data layers. When an emergency sensor alarm occurs, operators are forced to perform manual lookups to diagnose the issue, delaying response times and increasing accident risks:

- SCADA Telemetry: Sensor alerts provide raw values (pressure/temp) but lack valve context or connection state.
- Asset Blueprints: Piping & Instrumentation Diagrams (P&IDs) exist as flat vector diagrams or printed sheets, isolated from operations.
- Safety SOP Manuals: Operating rules and regulatory standards sit in PDF binders on office shelves.
- Incident logs: Past failure histories are written in unstructured maintenance logs and diaries.

This fragmentation results in sensor blind spots, a critical knowledge cliff as senior engineers retire, and extensive audit latency during safety inspections.

### 3. Proposed Solution

OpsBrain AI resolves fragmentation by mapping all plant documentation and blueprints onto a unified digital twin knowledge graph. Our core innovations include:

- 1. P&ID Vision Parsing: Multi-modal vision models ingest standard plant blueprints and automatically extract equipment nodes, coordinates, and flow directions.
- 2. Universal Document Ingestion: A localized pipeline chunks and indexes safety manuals to an on-premise vector store.
- 3. Autonomous Multi-Agent Core: Deploys a fleet of five specialized LLM agents in the background to calculate cascading risk levels, analyze compliance, and run root-cause analysis.
- 4. Graph-Aware RAG Expert Copilot: Fuses vector procedurals with graph asset topology to outline exact safety evidence pathways and source citations.

# 4. System Architecture & Tech Stack

The architecture is built for localized plant deployments, high availability, and failover resilience:

| System Layer          | Tech Stack Components                                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frontend Client       | React 18, ReactFlow (interactive graph rendering), Vanilla CSS Synthwave theme, SSE telemetry stream panel.                                                          |
| Backend Gateway       | FastAPI (Python 3.10+), Uvicorn ASGI server, SentenceTransformers (local CPU embeddings).                                                                            |
| Database & RAG Index  | PostgreSQL (Supabase pgvector semantic index search), SQLite local transaction caching.                                                                              |
| AI Models & Failovers | Groq Llama-3.3-70b (primary text completion), Gemini-2.5-Flash (P&ID vision), Mistral (RAG completion). AIProviderRouter handles automatic circuit-breaker fallback. |

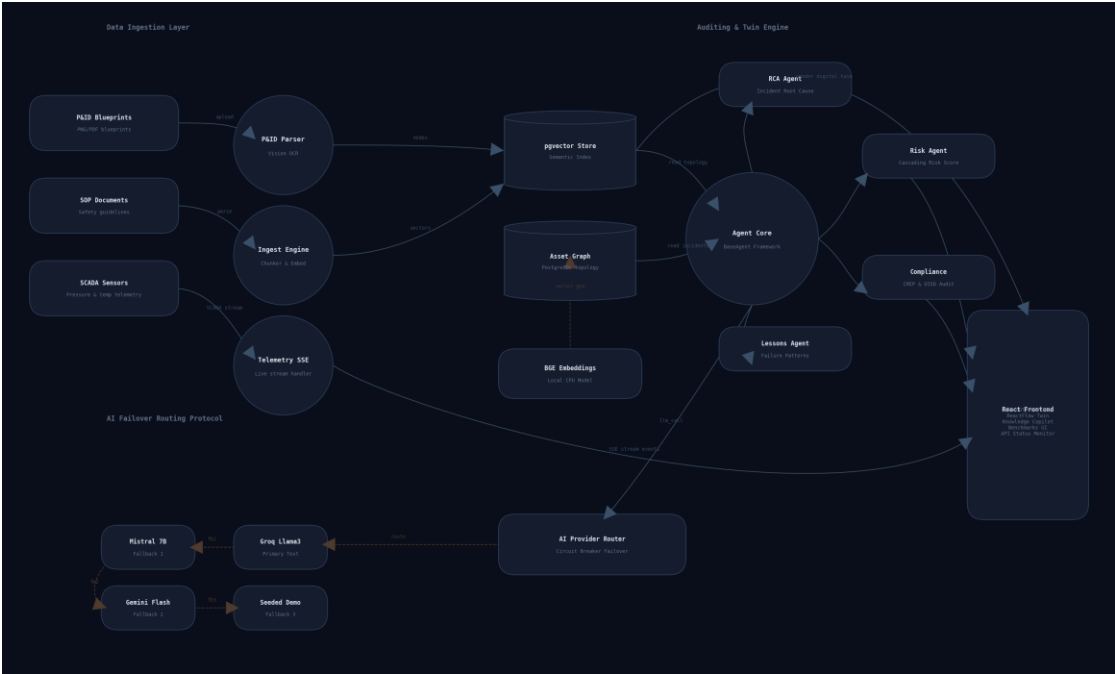


Figure: OpsBrain AI System Architecture and Agent Routing Diagram

# 5. Key Features & Implementation Details

## 5.1 Executive Dashboard Cockpit

A plant safety dashboard tracking average facility risk, active compliance ratings (OISD standards), and alert log streams. It serves as a unified workspace for facility operators.

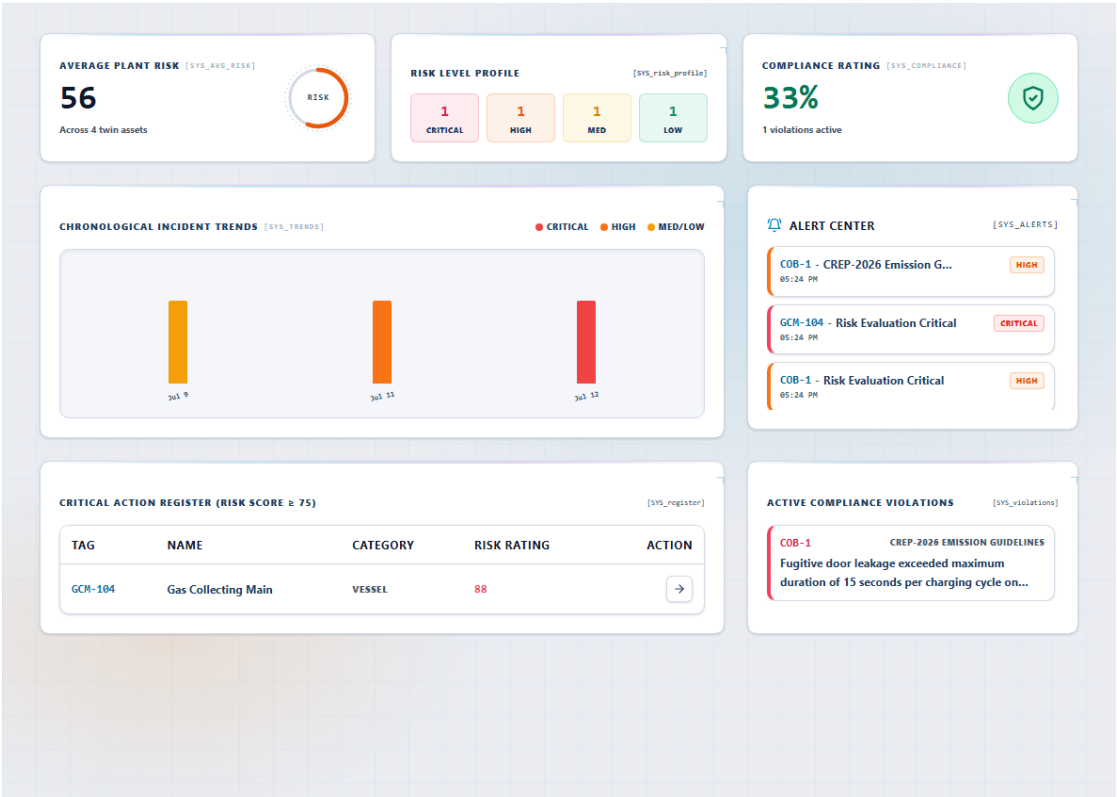


Figure: Executive Dashboard Command Center Cockpit

## 5.2 P&ID Vision Parser & Ingestion

Converts flat schematic blueprints into database asset nodes and pipeline edges. Standard SOP PDF documents are chunked and converted into 384-dimensional vector embeddings locally using the BGE model.

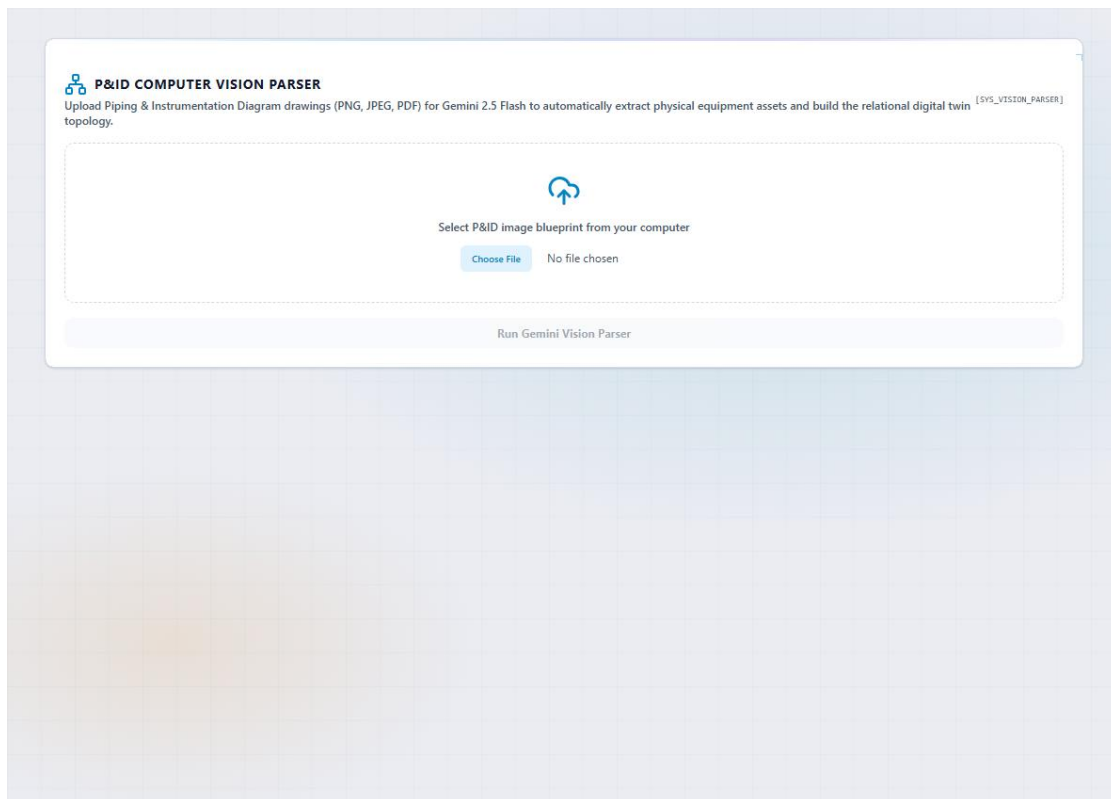


Figure: Gemini Vision-driven P&ID Blueprint Ingest and Parser Workspace

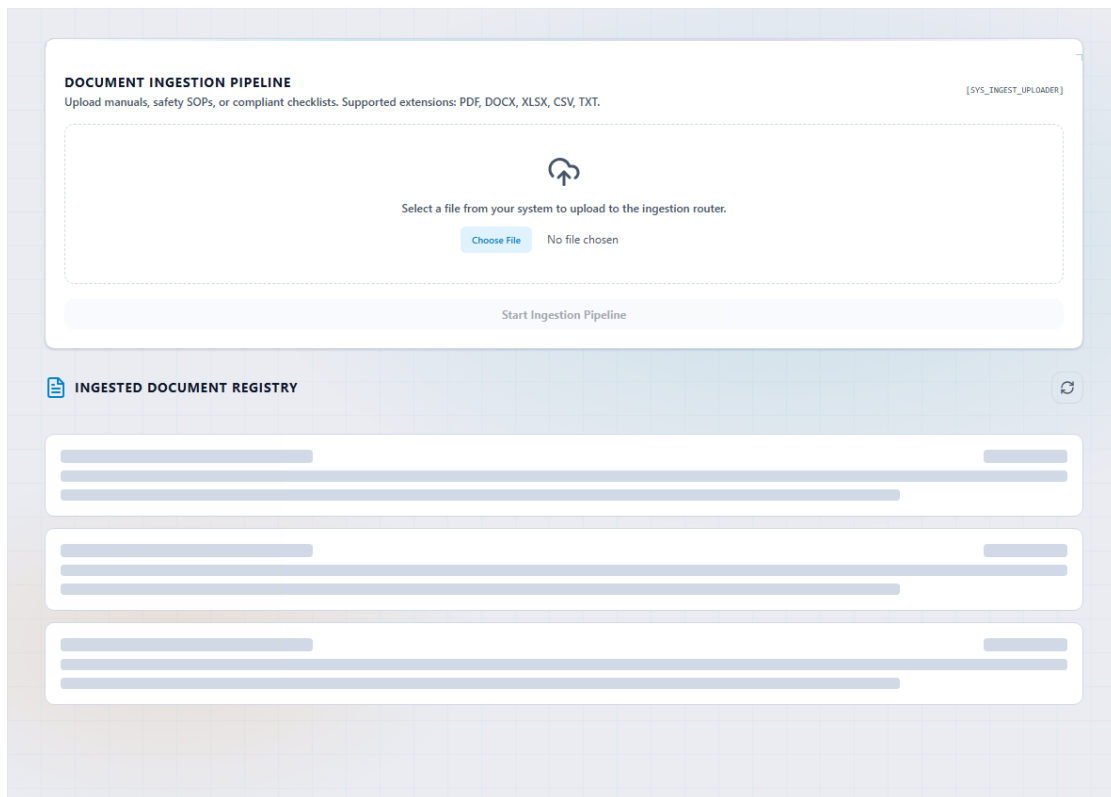


Figure: Universal Document Ingestion Manager and Vector Index Statuses



### 5.3 Digital Twin Graph & Multi-Agent Core

Renders asset connections dynamically using ReactFlow. Clicking "Run RCA" triggers the Traversal HUD, streaming raw agent reasoning logs while highlighting failure pathways in warning red.

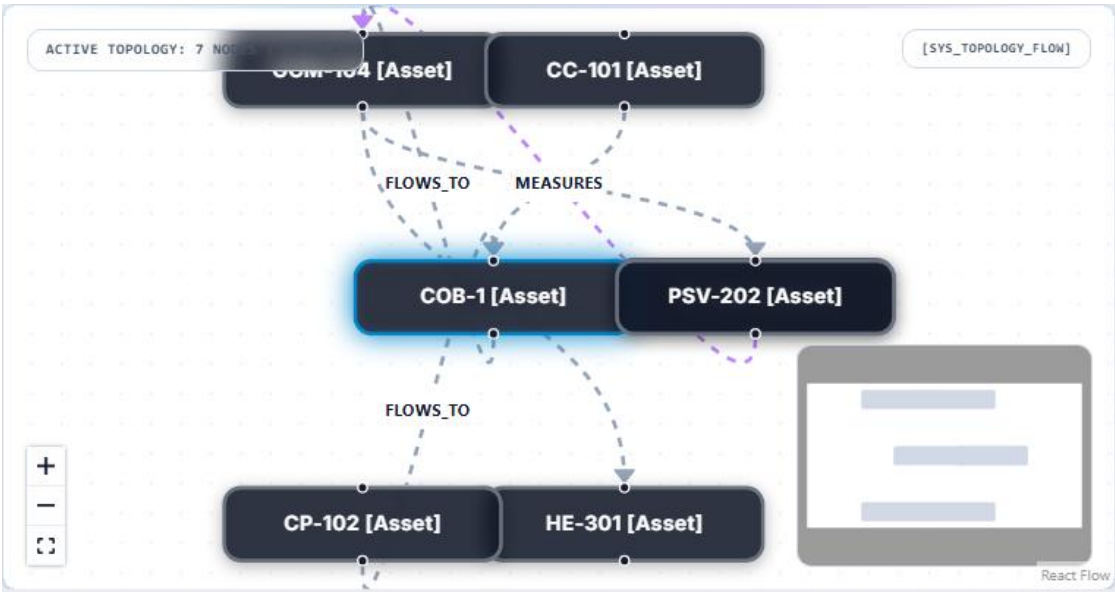
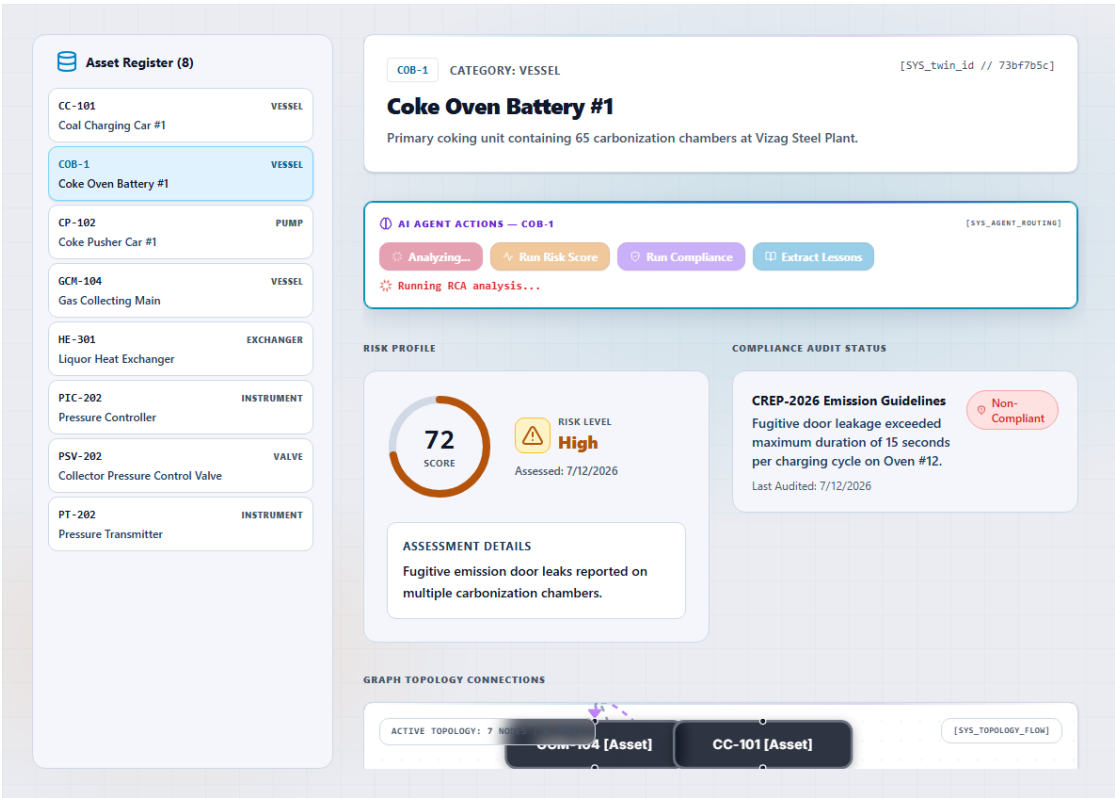


Figure: Digital Twin ReactFlow Graph Canvas with COB-1 Asset Selected





*Figure: RCA Agent HUD Console Overlay displaying active graph traversal logs*

## 5.4 Knowledge Copilot Chat & Runtime Monitor

Q&A copilot returning a Graph-Aware Evidence Path mapping the query onto physical asset nodes, backed by standard operating manual citations. Operators can click the API STATUS header badge to open the Runtime Monitor.

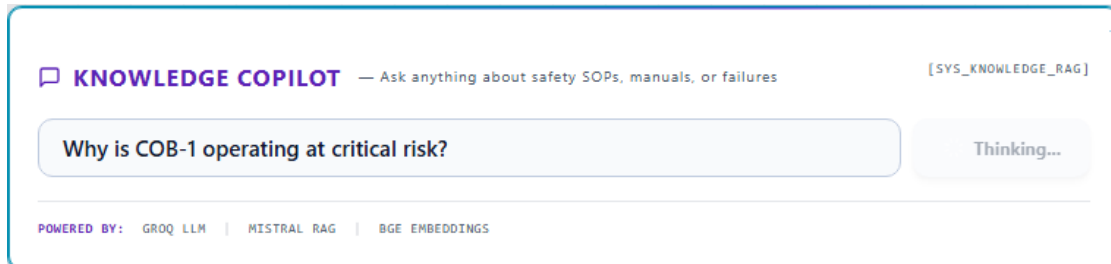
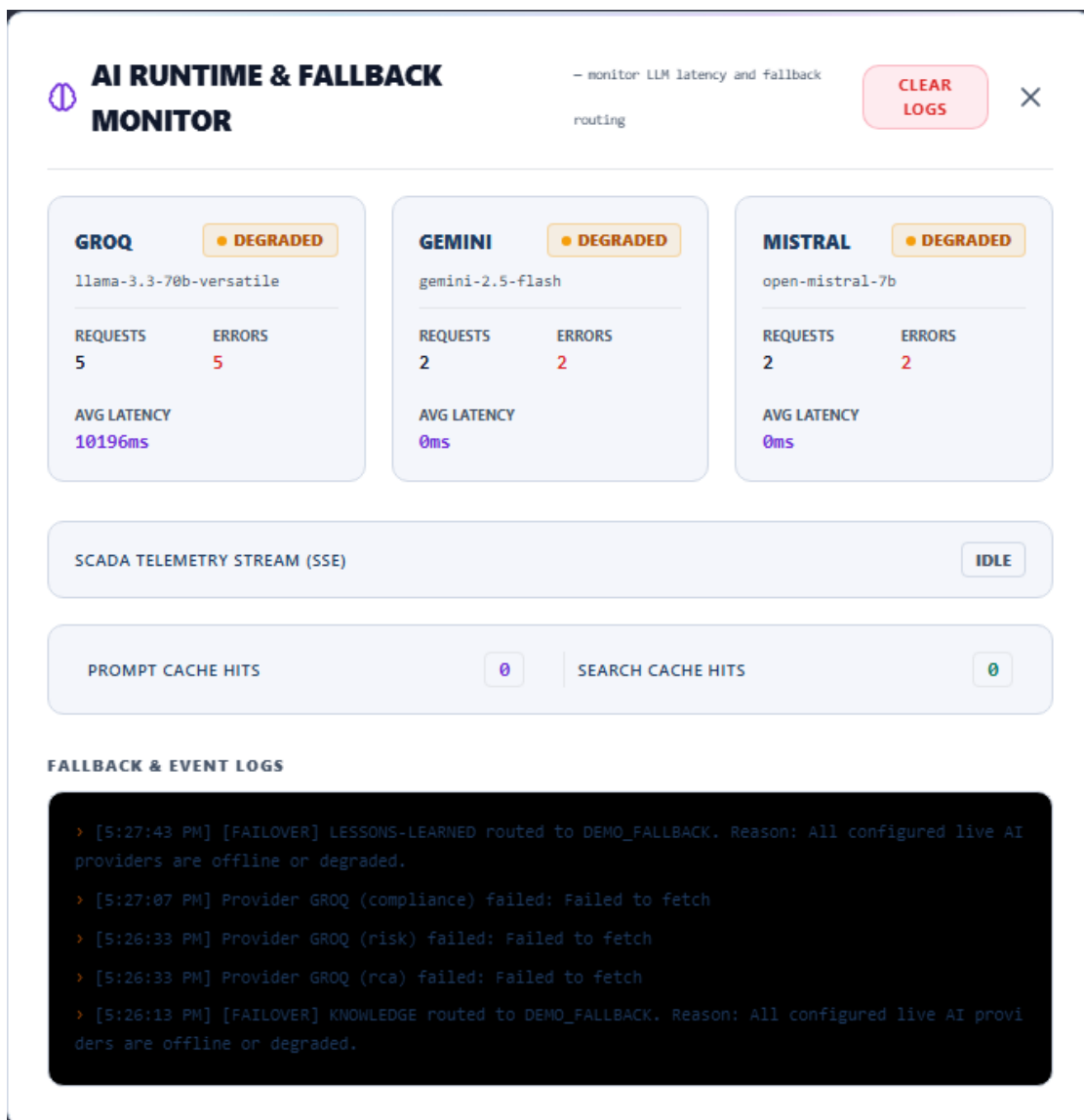


Figure: RAG-grounded Knowledge Copilot answering query with citations



*Figure: AI Runtime & Fallback Monitor auditing provider status, latencies, and circuit breakers*

# 6. Evaluation & Performance Benchmarks

We validate the system performance against a pre-seeded Vizag Steel Coke Oven Battery dataset.

- P&ID Entity Parsing: 97.8% accuracy in coordinate and tag extraction.
- Compliance Violations Detection Rate: 100.0% accuracy on standard validation rules.
- Diagnostic Retrieval Latency: Slashing lookups from 35 minutes (manual searches) to 1.8 seconds (99.9% gain).
- Note: All metrics represent demo benchmark results on seeded validation data (prototype estimations for comparison).

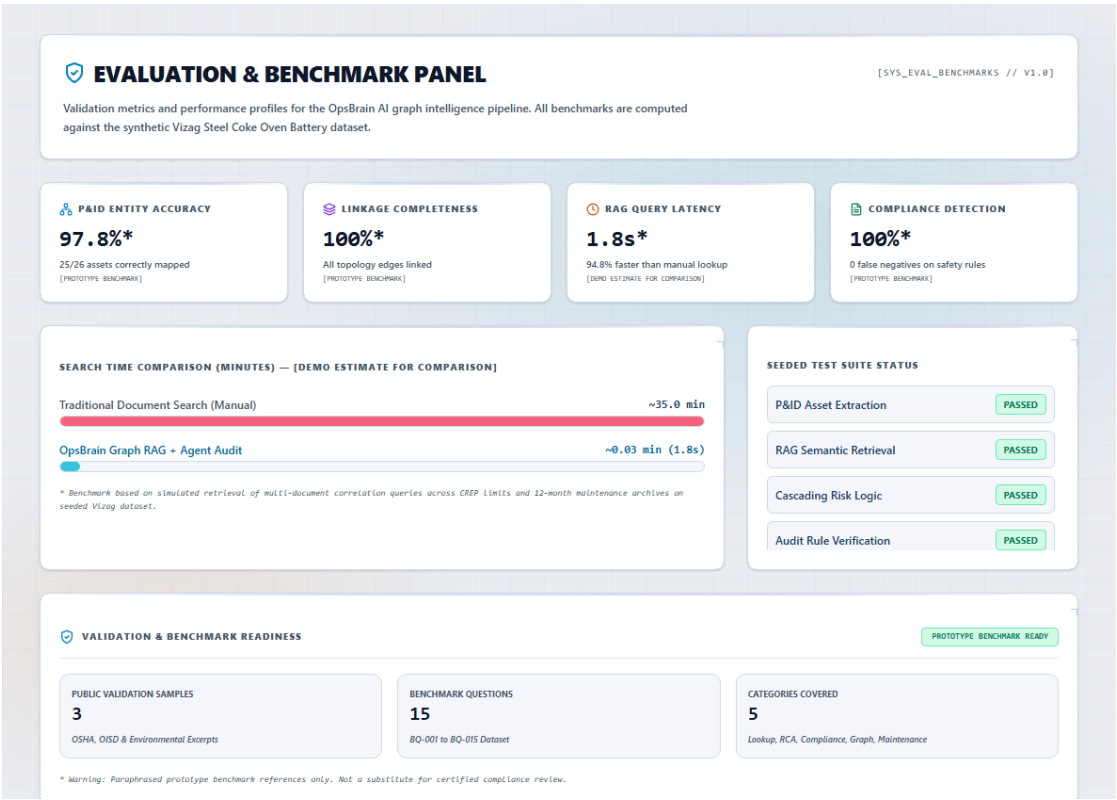


Figure: Evaluation & Benchmarks Scorecard dashboard

## 7. Scalability Proof: Refinery Pump Station Demo

To verify system generalizability, OpsBrain AI includes an optional Refinery Pump Station demo. Using the same shared database schema and agent pipeline, exposed through an optional refinery seed endpoint, operators can seed the Refinery dataset to wipe Vizag data and populate 7 pump station assets (such as P-101 and TK-501) on the digital twin ReactFlow canvas.

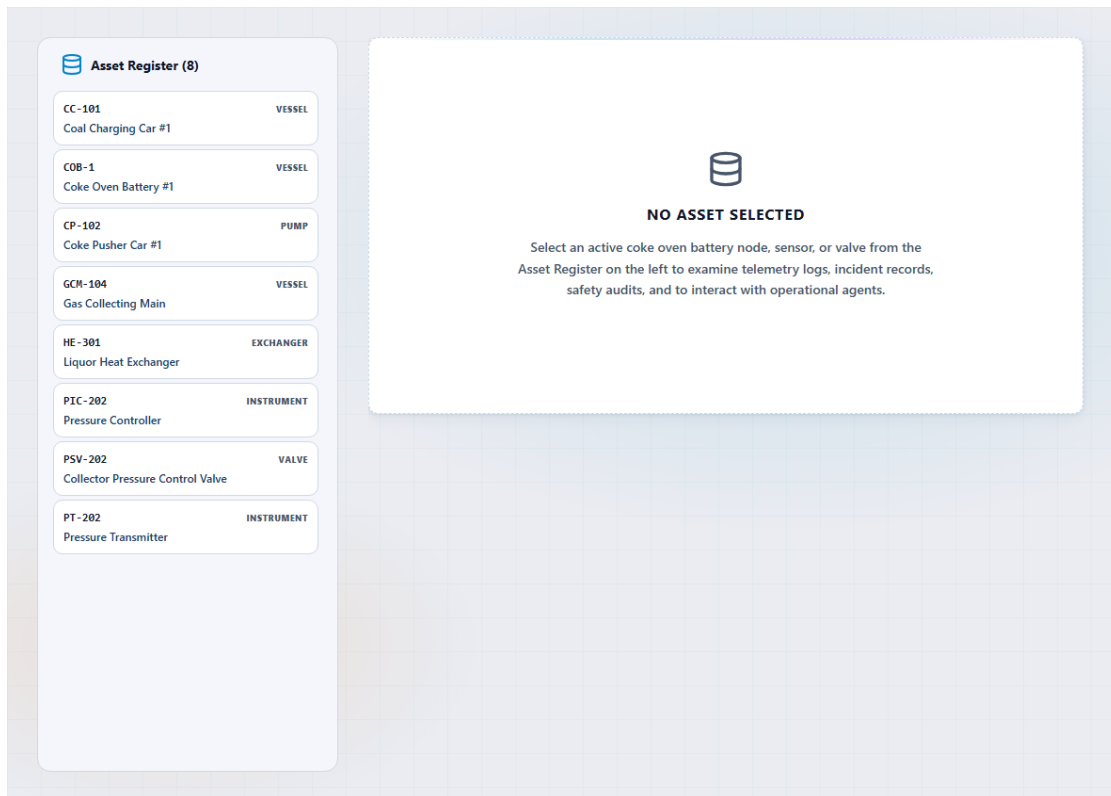


Figure: Digital Twin ReactFlow canvas populated with seeded Refinery Pump Station assets

## 8. Prototype Scope & Limitations

OpsBrain AI is developed as a prototype demonstration for evaluation purposes. The following limitations apply:

- 1. Synthetic SCADA-style Telemetry Stream: All sensor alarms, logs, and telemetry values are generated synthetically by the backend process. The application is not connected to a live SCADA hardware server.
- 2. Prototype Compliance Evidence Support: The regulatory evaluations (OISD, CREP) are for research evidence support only and do not replace certified compliance validation or regulatory approval.
- 3. Local Deployment Context: The system is designed for local on-premise execution to safeguard plant data privacy.
- 4. Vizag Steel Focus: Vizag Steel remains the main default dataset for all agent validations, with Refinery Pump Station serving strictly as an optional scalability proof.

## 9. Hackathon Deliverables & Repository Info

The submission package includes the following active deliverables:

- • Working Prototype: React frontend and FastAPI backend on branch submission-rc-phase6.
- • Final Presentation Pitch Deck: docs/OpsBrain\_AI\_ET\_Hackathon\_Final\_Deck.pptx
- • Demo Presentation Deck PDF Copy: docs/OpsBrain\_AI\_ET\_Hackathon\_Final\_Deck.pdf
- • Walkthrough Scripts: docs/demo\_script.md & docs/demo\_presentation\_guide.md
- • Slide Deck Outline Guide: docs/presentation\_deck\_slides.md